

InternVL

: Scaling up Vision Foundation Models and Aligning
for Generic Visual-Linguistic Tasks

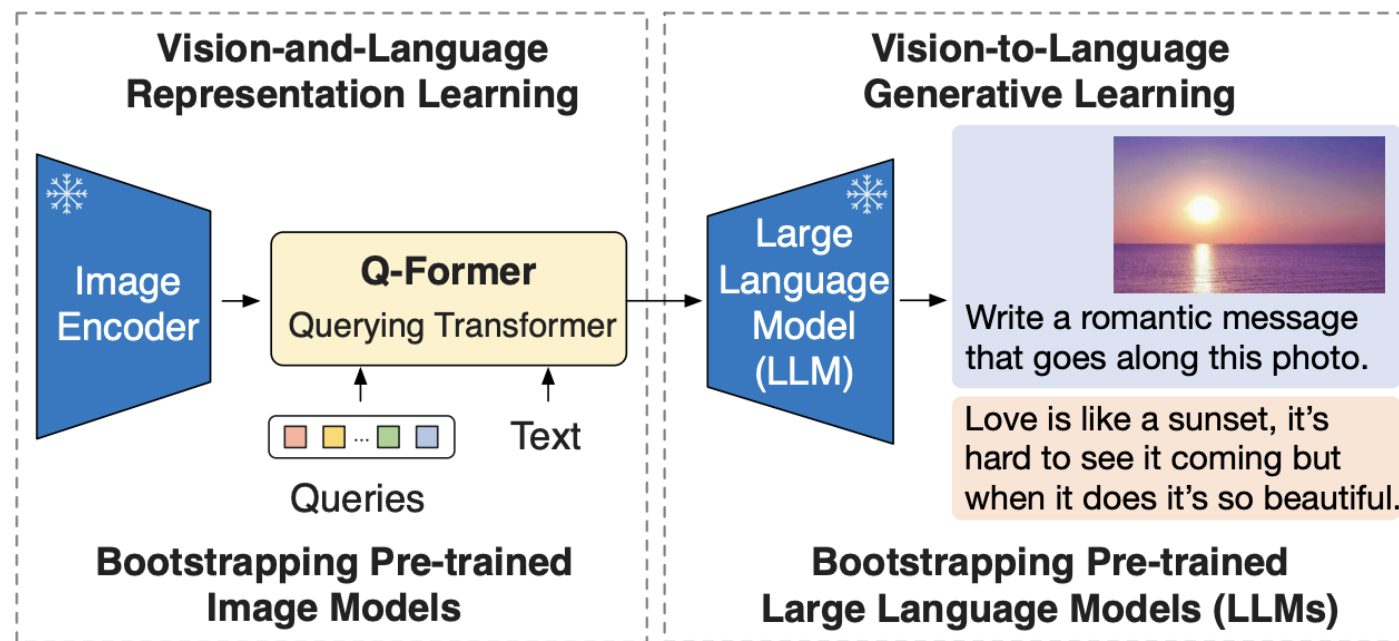
이미지 처리팀: 안가영(발표자), 황장훈

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1. Motivation
2. Model Design
3. Training
4. Experiments
5. Conclusion

BLIP-2

: Bootstrapping Language-Image Pre-training
with Frozen Image Encoders and Large Language Models



<BLIP-2 framework>

lightweight “glue” layers의 한계점

- **Disparity in parameter scale :**

LLM보다 vision encoder의 크기가 너무 작아 LLM 의 능력을 충분하게 사용 못할 수도 있음

- **Inconsistent representation :**

vision data로만 학습되거나 BERT계열 모델들은 종종 비전과 align이 잘 되지 않는 경우가 있음

- **Inefficient connection**

vision과 language를 align시키는 메커니즘이 경량화 및 랜덤 초기화되어 있어 multi modal 이해가 부족할 수 있음

InternVL의 핵심 설계 요소

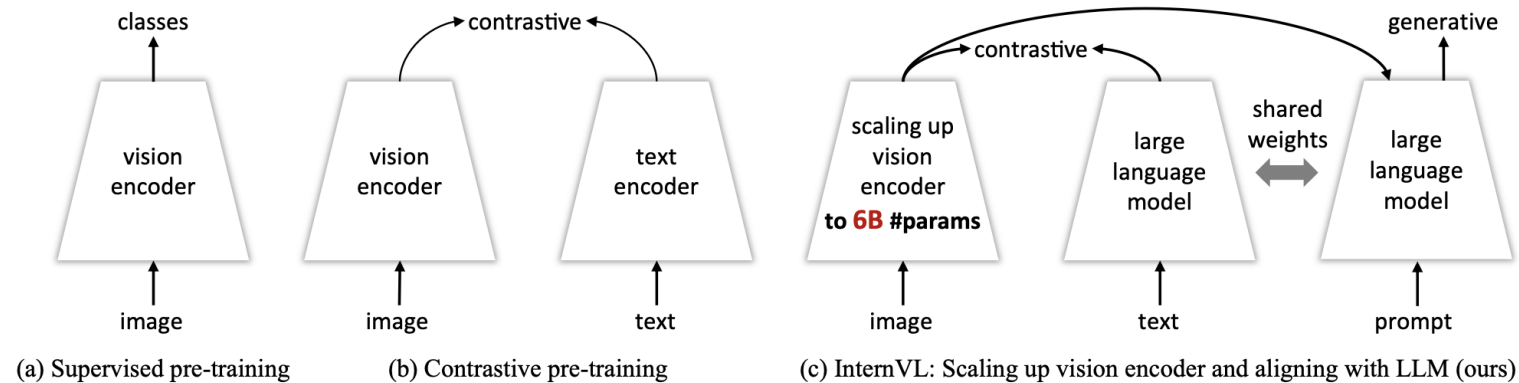
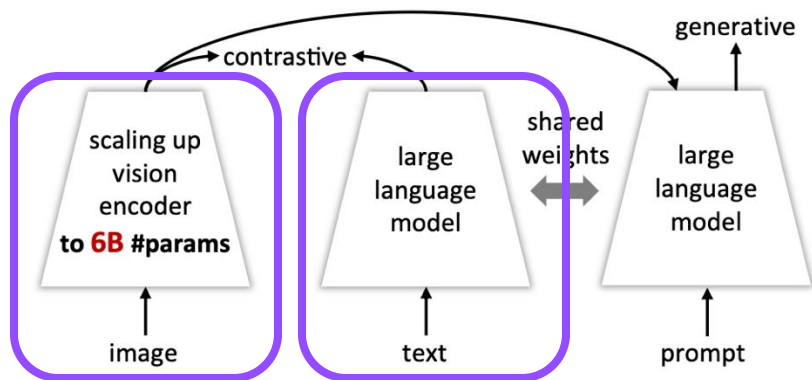


Figure 1. **Comparisons of different vision and vision-language foundation models.** (a) indicates the traditional vision foundation model, *e.g.* ResNet [57] pre-trained on classification tasks. (b) represents the vision-language foundation models, *e.g.* CLIP [117] pre-trained on image-text pairs. (c) is our InternVL, which presents a workable way to align the large-scale vision foundation model (*i.e.*, InternViT-6B) with the large language model and is versatile for both contrastive and generative tasks.

1. Parameter-balanced vision and language components
2. Consistent representations
3. Progressive image-text alignment

InternVL의 핵심 설계 요소

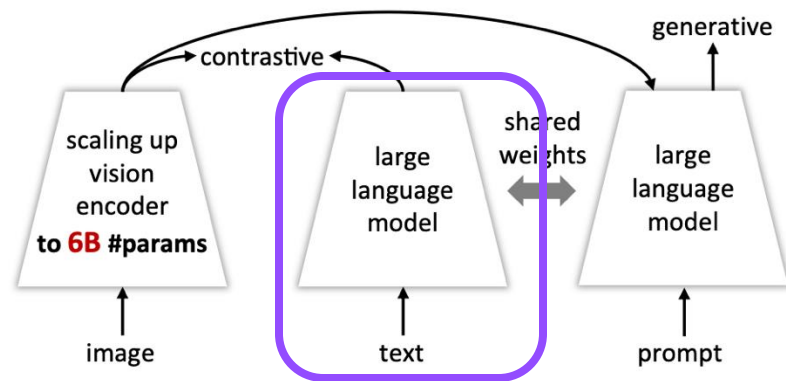


(c) InternVL: Scaling up vision encoder and aligning with LLM (ours)

1. Parameter-balanced vision and language components

- vision encoder의 파라미터 개수를 LLM과 비슷하게 60억개로 설정
- LLM middleware를 통해 유저 명령에 기반한 vision feature 재구성

InternVL의 핵심 설계 요소

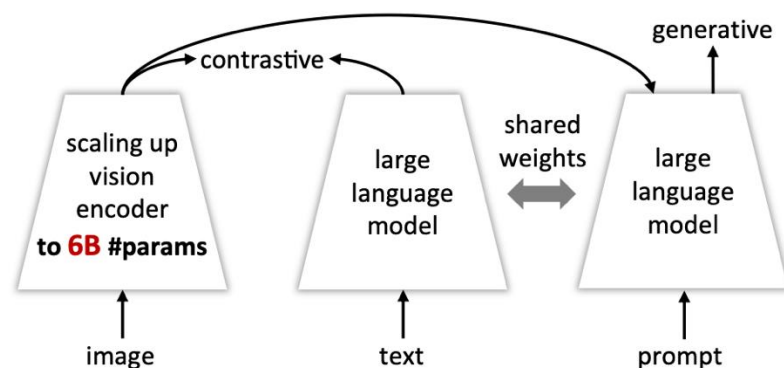


(c) InternVL: Scaling up vision encoder and aligning with LLM (ours)

2. Consistent representations

- LLM middleware를 multilingual LLaMA로 초기화

InternVL의 핵심 설계 요소

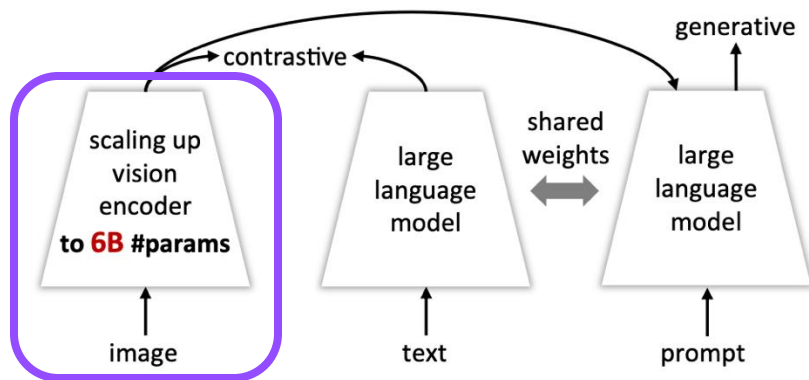


(c) InternVL: Scaling up vision encoder and aligning with LLM (ours)

3. Progressive image-text alignment

- 다양한 image-text data를 통해 학습 안정성 확보
- contrastive learning on large-scale noisy image-text data
- generative learning on fine-grained data

InternVL의 Vision Encoder



(c) InternVL: Scaling up vision encoder and aligning with LLM (ours)

1. Large-Scale Vision Encoder: InternViT-6B

- vanilla vision transformer(ViT)
- 6B parameter
- hyperparameter 탐색 수행
 - model depth {32, 48, 64, 80}
 - head dimension {64, 128}
 - MLP ratio {4, 8}

InternVL의 Vision Encoder

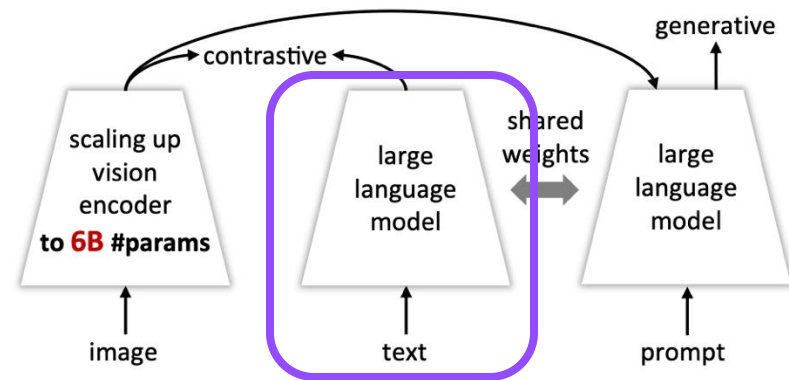
name	width	depth	MLP	#heads	#param (M)
ViT-G [173]	1664	48	8192	16	1843
ViT-e [23]	1792	56	15360	16	3926
EVA-02-ViT-E [130]	1792	64	15360	16	4400
ViT-6.5B [128]	4096	32	16384	32	6440
ViT-22B [37]	6144	48	24576	48	21743
InternViT-6B (ours)	3200	48	12800	25	5903

Table 1. Architecture details of the InternViT-6B model.

1. Large-Scale Vision Encoder: InternViT-6B

- 속도 : GPU 충분하면 depth 영향 없지만
한정적이라면 depth가 적을수록 빠름
- 정확도 : 파라미터 개수가 같으면
hyperparameter 조합에 따른 성능 큰 차이 없음
- 이를 반영하여 vision encoder인
InternViT-6B 구축

InternVL의 Middleware



(c) InternVL: Scaling up vision encoder and aligning with LLM (ours)

2. Language Middleware: QLLaMA

- pre-trained multilingual LLaMA-7B
- 96개의 learnable queries와 cross-attention layer를 포함시켜 parameter 1B를 추가하여 구성

InternVL의 Middleware

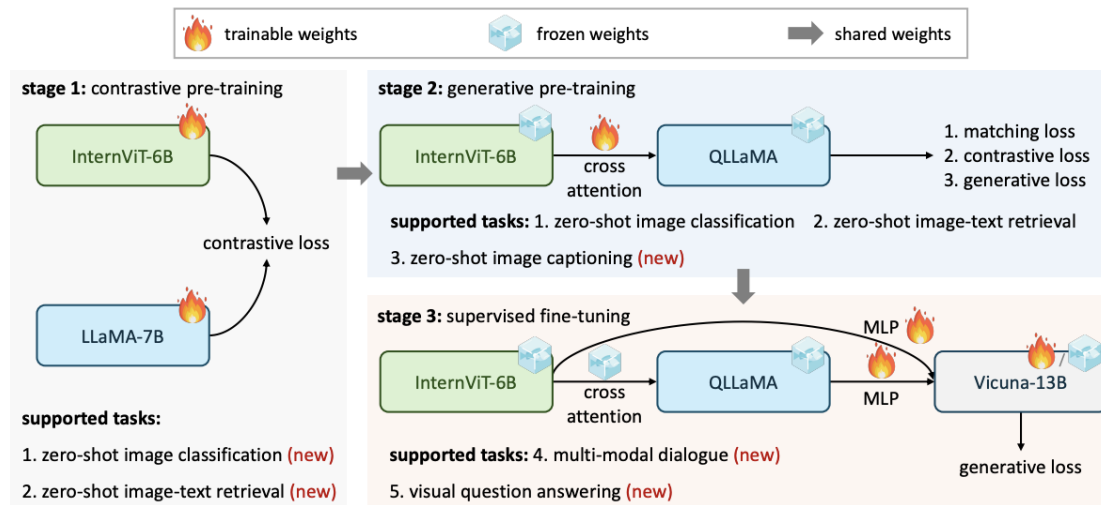
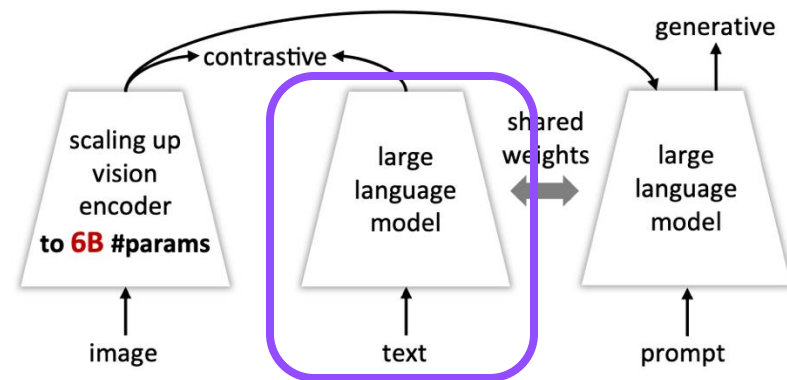


Figure 3. **The training strategy of the proposed InternVL model.** It consists of three progressive stages, including vision-language contrastive training, vision-language generative training, and supervised fine-tuning. These stages effectively leverage public data from diverse sources, ranging from noisy image-text pairs on the web to high-quality caption, VQA, and multi-modal dialogue datasets.

2. Language Middleware: QLLaMA

- pre-trained multilingual LLaMA-7B과 InternViT를 학습
- 이렇게 파인튜닝된 LLaMA를 기반으로 96개의 learnable queries와 cross-attention layer를 포함시켜 parameter 1B를 추가하여 구성

InternVL의 Middleware

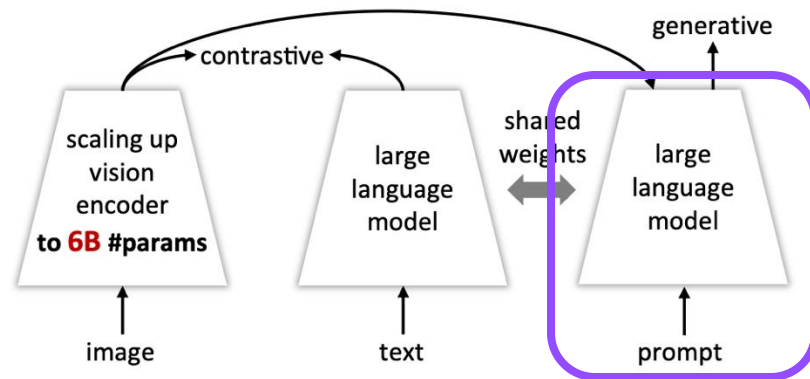


(c) InternVL: Scaling up vision encoder and aligning with LLM (ours)

2. Language Middleware: QLLaMA

- InternViT-6B가 생성한 이미지 토큰을 LLM과 정렬된 표현으로 변환
- LLM 디코더가 고정되어 있어도 InternVL은 멀티모달 대화 task에서 좋은 성능 보임
- contrastive learning 적용 가능

InternVL의 LLM



(c) InternVL: Scaling up vision encoder and aligning with LLM (ours)

3. Language Decoder

- 상용 LLM 사용 가능
- 논문에서는 Vicuna-13B 선택

InternVL "Swiss Army Knife" Model

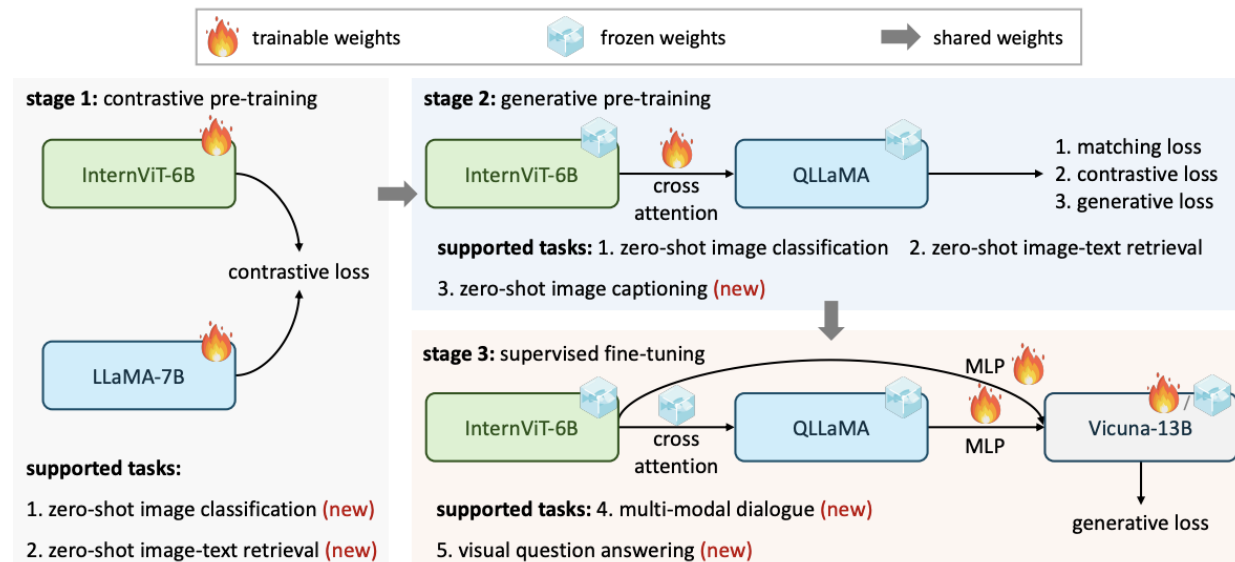


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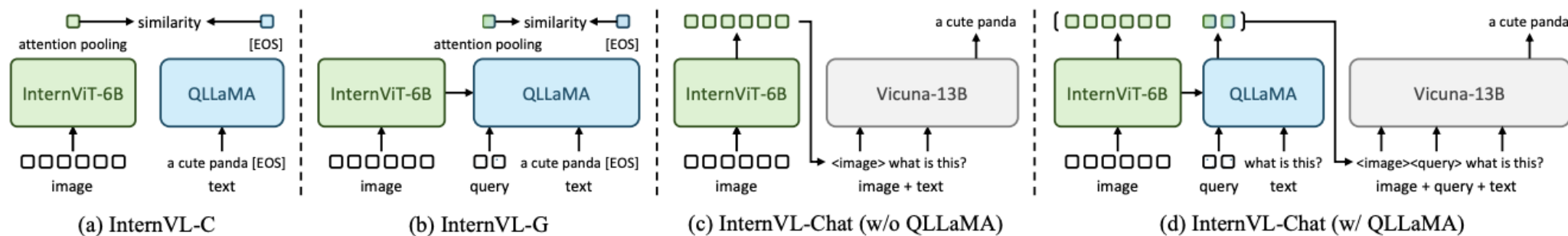
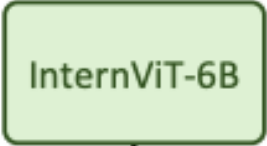


Figure 4. **Different ways to use InternVL.** By flexibly combining the vision encoder and the language middleware, InternVL can support various vision-language tasks, including contrastive tasks, generative tasks, and multi-modal dialogue.

InternVL “Swiss Army Knife” Model



InternViT-6B

1. Visual Perception Task

- Only InternViT로 수행 가능
- Feature map 생성
- global average pooling or linear projection을 통한 이미지 분류

InternVL "Swiss Army Knife" Model

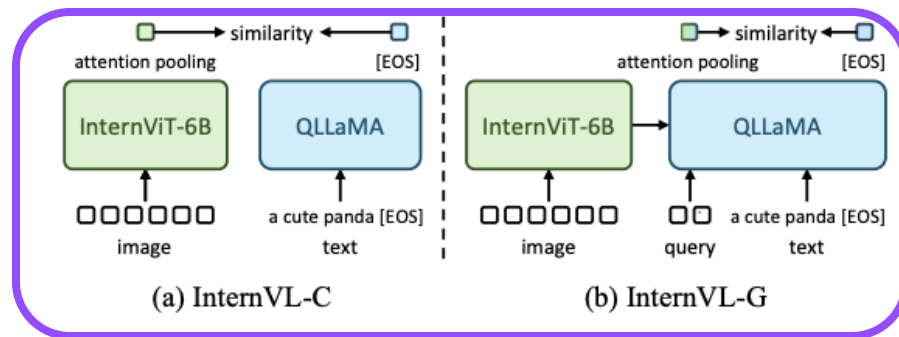


Figure 4. **Different ways to use InternVL.** By flexibly c various vision-language tasks, including contrastive tasks

2. contrastive tasks

- InternVL-C(contrastive)와 InternVL-G(generative)를 도입
- global visual feature (I_f)계산
 - InternVL-C는 InternViT의 output인 image feature,
InternVL-G는 InternViT의 output인 image feature와
QLLaMA에서 learnable queries와 cross-attention 적용한
query feature에
attention pooling을 적용하여 global visual feature를 계산

InternVL "Swiss Army Knife" Model

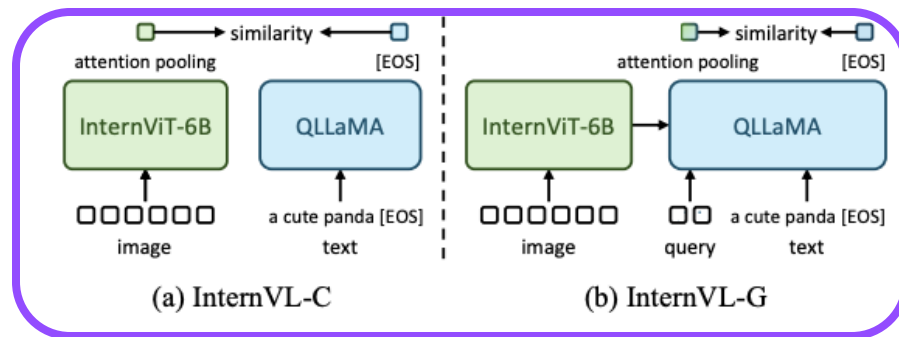


Figure 4. **Different ways to use InternVL.** By flexibly combining various vision-language tasks, including contrastive tasks

2. contrastive tasks

- InternVL-C(contrastive)와 InternVL-G(generative)를 도입
- global text feature(T_f) 계산
 - QLLaMA의 EOS 토큰에서 입력 text 특징을 추출
- I_f 와 T_f 간의 유사도 연산하여 contrastive task 수행

InternVL “Swiss Army Knife” Model

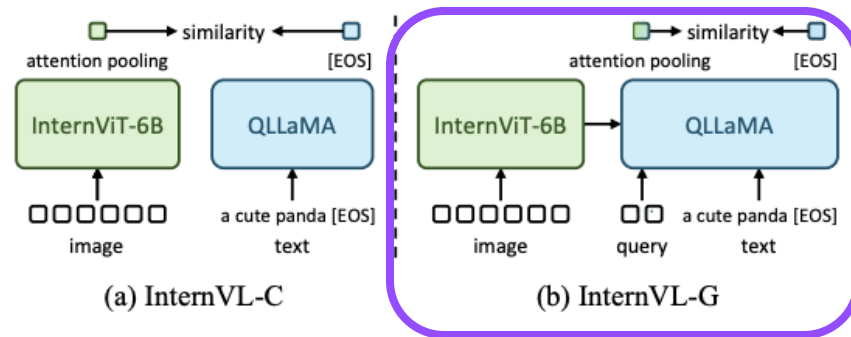
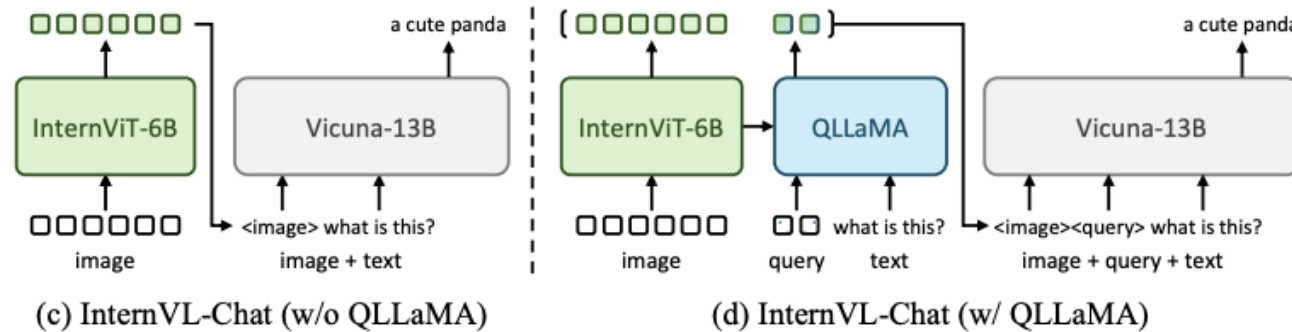


Figure 4. **Different ways to use InternVL.** By flexibly c various vision-language tasks, including contrastive tasks

3. generation tasks

- scale up 되어있는 QLLaMA를 통해 image captioning task에서 좋은 성능을 보임

InternVL “Swiss Army Knife” Model



4. multi-modal dialogue

- (c) : InternViT-6B를 독립적으로 사용
- (d) : 완전한 InternVL 모델을 동시에 사용
- (d)는 (c)보다 복잡한 multimodal task 가능

Training

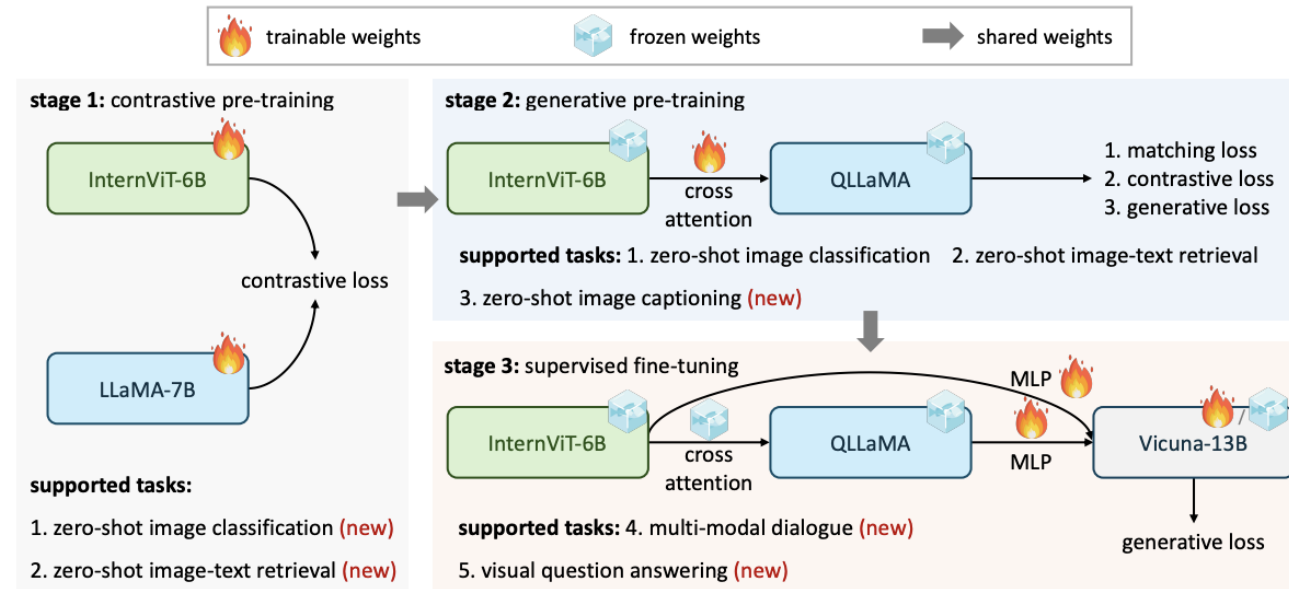
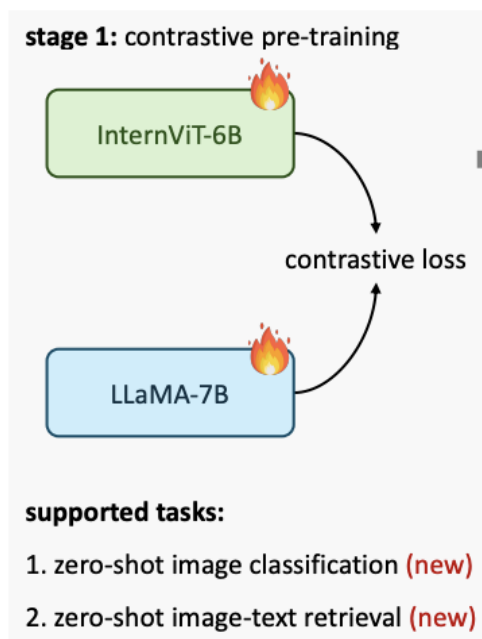


Figure 3. **The training strategy of the proposed InternVL model.** It consists of three progressive stages, including vision-language contrastive training, vision-language generative training, and supervised fine-tuning. These stages effectively leverage public data from diverse sources, ranging from noisy image-text pairs on the web to high-quality caption, VQA, and multi-modal dialogue datasets.

Vision-Language Contrastive Training



- InternViT-6B와 multilingual LLaMA-7B 대조 학습으로 정렬
- CLIP 목표함수 사용
: 배치 내 image-text pair의 유사도에 대한 cross-entropy 손실함수 사용
- Contrastive task에 특화
: zero-shot image classification, image-text retrieval 등
- 사용된 dataset
: LAION-en , LAION-multi , LAION-COCO , COYO , Wukong 등

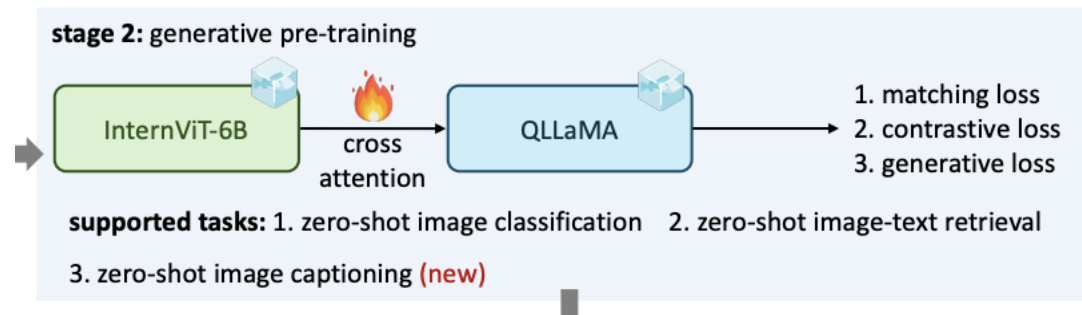
Vision-Language Contrastive Training

dataset	characteristics		stage 1		stage 2	
	language	original	cleaned	remain	cleaned	remain
LAION-en [120]	English	2.3B	1.94B	84.3%	91M	4.0%
LAION-COCO [121]		663M	550M	83.0%	550M	83.0%
COYO [14]		747M	535M	71.6%	200M	26.8%
CC12M [20]		12.4M	11.1M	89.5%	11.1M	89.5%
CC3M [124]		3.0M	2.6M	86.7%	2.6M	86.7%
SBU [112]		1.0M	1.0M	100%	1.0M	100%
Wukong [55]	Chinese	100M	69.4M	69.4%	69.4M	69.4%
LAION-multi [120]	Multi	2.2B	1.87B	85.0%	100M	4.5%
Total	Multi	6.03B	4.98B	82.6%	1.03B	17.0%

Table 2. **Details of the training data for InternVL in stage 1 and stage 2.** Among them, LAION-en [120], LAION-multi [120], COYO [14], and Wukong [55] are web-scale image-text pairs data. LAION-COCO [121] is a synthetic dataset with high-quality captions from LAION-en. CC12M [20], CC3M [124], SBU [112] are academic caption datasets. “Multi” means multilingual.

- 저품질 데이터 일부 제외
- image-text pair dataset 기준
6.03B에서 4.98B dataset으로 학습

Vision-Language Generative Training (Stage 2)



- InternViT-6B와 QLLaMA를 연결
 - InternViT-6B와 QLLaMA를 frozen
- 새롭게 추가된 learnable queries와 cross-attention 레이어만을 학습

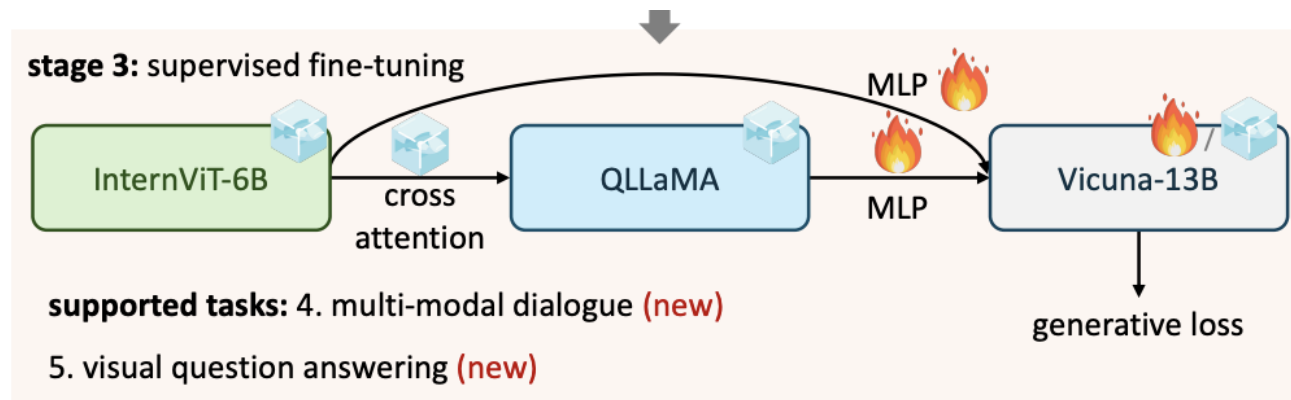
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- 저품질 데이터 일부 추가적으로 제외
- image-text pair dataset 기준
6.03B에서 1.03B dataset으로 학습

Supervised Fine-tuning (Stage 3)



- 상용화된 LLM decoder(Vicuna)와 MLP연결하여 SFT 수행
- LLM decoder는 freeze하거나 함께 학습할 수 있음

Supervised Fine-tuning (Stage 3)

task	#samples	dataset
Captioning	588K	COCO Caption [22], TextCaps [126]
VQA	1.1M	VQAv2 [54], OKVQA [104], A-OKVQA [122], IconQA [99], AI2D [71], GQA [64]
OCR	294K	OCR-VQA [107], ChartQA [105], DocVQA [29], ST-VQA [12], EST-VQA [150], InfoVQA [106], LLaVAR [182]
Grounding	323K	RefCOCO+/g [103, 170], Toloka [140]
Grounded Cap.	284K	RefCOCO+/g [103, 170]
Conversation	1.4M	LLaVA-150K [92], SVIT [183], VisDial [36], LRV-Instruction [90], LLaVA-Mix-665K [91]

Table 3. Details of the training data for InternVL in stage 3.

We collect a wide range of high-quality instruction data, totaling approximately 4 million samples. For a fair comparison, we only use the training split of these datasets.

- non-dialogue dataset인 경우, LLaVA를 통해 dataset 변환하여 구축

Visual Perception Benchmarks : Transfer to Image Classification

method	#param	IN-1K	IN-ReaL	IN-V2	IN-A	IN-R	IN-Ske	avg.
OpenCLIP-H [67]	0.6B	84.4	88.4	75.5	—	—	—	—
OpenCLIP-G [67]	1.8B	86.2	89.4	77.2	63.8	87.8	66.4	78.5
DINOv2-g [111]	1.1B	86.5	89.6	78.4	75.9	78.8	62.5	78.6
EVA-01-CLIP-g [46]	1.1B	86.5	89.3	77.4	70.5	87.7	63.1	79.1
MAWS-ViT-6.5B [128]	6.5B	87.8	—	—	—	—	—	—
ViT-22B* [37]	21.7B	89.5	90.9	83.2	83.8	87.4	—	—
InternViT-6B (ours)	5.9B	88.2	90.4	79.9	77.5	89.8	69.1	82.5

Table 4. **Linear evaluation on image classification.** We report the top-1 accuracy on ImageNet-1K [38] and its variants [10, 60, 61, 119, 141]. * ViT-22B [37] uses the private JFT-3B dataset [173].

Visual Perception Benchmarks : Transfer to Semantic Segmentation

method	#param	crop size	1/16	1/8	1/4	1/2	1
ViT-L [137]	0.3B	504^2	36.1	41.3	45.6	48.4	51.9
ViT-G [173]	1.8B	504^2	42.4	47.0	50.2	52.4	55.6
ViT-22B [37]	21.7B	504^2	44.7	47.2	50.6	52.5	54.9
InternViT-6B (ours)	5.9B	504^2	46.5	50.0	53.3	55.8	57.2

(a) Few-shot semantic segmentation with limited training data. Following ViT-22B [37], we fine-tune the InternViT-6B with a linear classifier.

method	decoder	#param (train/total)	crop size	mIoU
OpenCLIP-G _{frozen} [67]	Linear	0.3M / 1.8B	512^2	39.3
ViT-22B _{frozen} [37]	Linear	0.9M / 21.7B	504^2	34.6
InternViT-6B _{frozen} (ours)	Linear	0.5M / 5.9B	504^2	47.2
ViT-22B _{frozen} [37]	UperNet	0.8B / 22.5B	504^2	52.7
InternViT-6B _{frozen} (ours)	UperNet	0.4B / 6.3B	504^2	54.9
ViT-22B [37]	UperNet	22.5B / 22.5B	504^2	55.3
InternViT-6B (ours)	UperNet	6.3B / 6.3B	504^2	58.9

(b) Semantic segmentation performance in three different settings, from top to bottom: linear probing, head tuning, and full-parameter tuning.

Table 5. **Semantic segmentation on ADE20K.** Results show that InternViT-6B has better pixel-level perceptual capacity.

Vision-Language Benchmarks : Zero-Shot Image-Text Retrieval

method	multi- lingual	Flickr30K (English, 1K test set) [116]						COCO (English, 5K test set) [22]						avg.
		Image → Text			Text → Image			Image → Text			Text → Image			
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	
Florence [171]	×	90.9	99.1	—	76.7	93.6	—	64.7	85.9	—	47.2	71.4	—	—
ONE-PEACE [143]	×	90.9	98.8	99.8	77.2	93.5	96.2	64.7	86.0	91.9	48.0	71.5	79.6	83.2
OpenCLIP-H [67]	×	90.8	99.3	99.7	77.8	94.1	96.6	66.0	86.1	91.9	49.5	73.4	81.5	83.9
OpenCLIP-g [67]	×	91.4	99.2	99.6	77.7	94.1	96.9	66.4	86.0	91.8	48.8	73.3	81.5	83.9
OpenCLIP-XLM-R-H [67]	✓	91.8	99.4	99.8	77.8	94.1	96.5	65.9	86.2	92.2	49.3	73.2	81.5	84.0
EVA-01-CLIP-g+ [130]	×	91.6	99.3	99.8	78.9	94.5	96.9	68.2	87.5	92.5	50.3	74.0	82.1	84.6
CoCa [169]	×	92.5	99.5	99.9	80.4	95.7	97.7	66.3	86.2	91.8	51.2	74.2	82.0	84.8
OpenCLIP-G [67]	×	92.9	99.3	99.8	79.5	95.0	97.1	67.3	86.9	92.6	51.4	74.9	83.0	85.0
EVA-02-CLIP-E+ [130]	×	93.9	99.4	99.8	78.8	94.2	96.8	68.8	87.8	92.8	51.1	75.0	82.7	85.1
BLIP-2 [†] [81]	×	97.6	100.0	100.0	89.7	98.1	98.9	—	—	—	—	—	—	—
InternVL-C (ours)	✓	94.7	99.6	99.9	81.7	96.0	98.2	70.6	89.0	93.5	54.1	77.3	84.6	86.6
InternVL-G (ours)	✓	95.7	99.7	99.9	85.0	97.0	98.6	74.9	91.3	95.2	58.6	81.3	88.0	88.8

method		Flickr30K-CN (Chinese, 1K test set) [77]						COCO-CN (Chinese, 1K test set) [84]						avg.
WuKong-ViT-L [55]	×	76.1	94.8	97.5	51.7	78.9	86.3	55.2	81.0	90.6	53.4	80.2	90.1	78.0
R2D2-ViT-L [159]	×	77.6	96.7	98.9	60.9	86.8	92.7	63.3	89.3	95.7	56.4	85.0	93.1	83.0
Taiyi-CLIP-ViT-H [176]	×	—	—	—	—	—	—	—	—	—	60.0	84.0	93.3	—
AltCLIP-ViT-H [26]	✓	88.9	98.5	99.5	74.5	92.0	95.5	—	—	—	—	—	—	—
CN-CLIP-ViT-H [162]	×	81.6	97.5	98.8	71.2	91.4	95.5	63.0	86.6	92.9	69.2	89.9	96.1	86.1
OpenCLIP-XLM-R-H [67]	✓	86.1	97.5	99.2	71.0	90.5	94.9	70.0	91.5	97.0	66.1	90.8	96.0	87.6
InternVL-C (ours)	✓	90.3	98.8	99.7	75.1	92.9	96.4	68.8	92.0	96.7	68.9	91.9	96.5	89.0
InternVL-G (ours)	✓	92.9	99.4	99.8	77.7	94.8	97.3	71.4	93.9	97.7	73.8	94.4	98.1	90.9

Table 7. **Comparison of zero-shot image-text retrieval performance.** We evaluate the retrieval capability in English using the Flickr30K [116] and COCO [22], as well as in Chinese using Flickr30K-CN [77] and COCO-CN [84]. [†]BLIP-2 [81] is finetuned on COCO and zero-shot transferred to Flickr30K, contributing to the enhanced zero-shot performance on Flickr30K.

Vision-Language Benchmarks : Zero-Shot Image Captioning

method	glue layer	LLM decoder	COCO	Flickr30K	NoCaps
Flamingo-9B [3]	Cross-Attn	Chinchilla-7B	79.4	61.5	–
Flamingo-80B [3]	Cross-Attn	Chinchilla-70B	84.3	67.2	–
KOSMOS-2 [115]	Linear	KOSMOS-1	–	66.7	–
PaLI-X-55B [24]	Linear	UL2-32B	–	–	126.3
BLIP-2 [81]	QFormer	Vicuna-13B	–	71.6	103.9
InstructBLIP [34]	QFormer	Vicuna-13B	–	82.8	121.9
Shikra-13B [21]	Linear	Vicuna-13B	–	73.9	–
ASM [149]	QFormer	Husky-7B	–	87.7	117.2
Qwen-VL [5]	VL-Adapter	Qwen-7B	–	85.8	121.4
Qwen-VL-Chat [5]	VL-Adapter	Qwen-7B	–	81.0	120.2
Emu [131]	QFormer	LLaMA-13B	112.4	–	–
Emu-I [131]	QFormer	LLaMA-13B	117.7	–	–
DreamLLM [41]	Linear	Vicuna-7B	115.4	–	–
InternVL-G (ours)	Cross-Attn	QLLaMA	128.2	79.2	113.7

Table 10. **Comparison of zero-shot image captioning.** QLLaMA inherently possesses promising zero-shot captioning capabilities thanks to its scaled-up parameters and datasets.

Vision-Language Benchmarks : Zero-Shot Image Captioning

method	visual encoder	glue layer	LLM	Res.	PT	SFT	train. param	image captioning			visual question answering				dialogue	
								COCO	Flickr	NoCaps	VQA ^{v2}	GQA	VizWiz	VQA ^T	MME	POPE
InstructBLIP [34]	EVA-g	QFormer	Vicuna-7B	224	129M	1.2M	188M	—	82.4	123.1	—	49.2	34.5	50.1	—	—
BLIP-2 [81]	EVA-g	QFormer	Vicuna-13B	224	129M	—	188M	—	71.6	103.9	41.0	41.0	19.6	42.5	1293.8	85.3
InstructBLIP [34]	EVA-g	QFormer	Vicuna-13B	224	129M	1.2M	188M	—	82.8	121.9	—	49.5	33.4	50.7	1212.8	78.9
InternVL-Chat (ours)	IViT-6B	QLLaMA	Vicuna-7B	224	1.0B	4.0M	64M	141.4*	89.7	120.5	72.3*	57.7*	44.5	42.1	1298.5	85.2
InternVL-Chat (ours)	IViT-6B	QLLaMA	Vicuna-13B	224	1.0B	4.0M	90M	142.4*	89.9	123.1	71.7*	59.5*	54.0	49.1	1317.2	85.4
Shikra [21]	CLIP-L	Linear	Vicuna-13B	224	600K	5.5M	7B	117.5*	73.9	—	77.4*	—	—	—	—	—
IDEFICS-80B [66]	CLIP-H	Cross-Attn	LLaMA-65B	224	1.6B	—	15B	91.8*	53.7	65.0	60.0	45.2	36.0	30.9	—	—
IDEFICS-80B-I [66]	CLIP-H	Cross-Attn	LLaMA-65B	224	353M	6.7M	15B	117.2*	65.3	104.5	37.4	—	26.0	—	—	—
Qwen-VL [5]	CLIP-G	VL-Adapter	Qwen-7B	448	1.4B [†]	50M [†]	9.6B	—	85.8	121.4	78.8*	59.3*	35.2	63.8	—	—
Qwen-VL-Chat [5]	CLIP-G	VL-Adapter	Qwen-7B	448	1.4B [†]	50M [†]	9.6B	—	81.0	120.2	78.2*	57.5*	38.9	61.5	1487.5	—
LLaVA-1.5 [91]	CLIP-L ₃₃₆	MLP	Vicuna-7B	336	558K	665K	7B	—	—	—	78.5*	62.0*	50.0	58.2	1510.7	85.9
LLaVA-1.5 [91]	CLIP-L ₃₃₆	MLP	Vicuna-13B	336	558K	665K	13B	—	—	—	80.0*	63.3*	53.6	61.3	1531.3	85.9
InternVL-Chat (ours)	IViT-6B	MLP	Vicuna-7B	336	558K	665K	7B	—	—	—	79.3*	62.9*	52.5	57.0	1525.1	86.4
InternVL-Chat (ours)	IViT-6B	MLP	Vicuna-13B	336	558K	665K	13B	—	—	—	80.2*	63.9*	54.6	58.7	1546.9	87.1
InternVL-Chat (ours)	IViT-6B	QLLaMA	Vicuna-13B	336	1.0B	4.0M	13B	146.2*	92.2	126.2	81.2*	66.6*	58.5	61.5	1586.4	87.6

Table 9. **Comparison with SoTA methods on 9 benchmarks.** Image captioning datasets include: COCO Karpathy test [22], Flickr30K Karpathy test [116], NoCaps val [2]. VQA datasets include: VQAv2 test-dev [54], GQA test-balanced [64], VizWiz test-dev [56], and TextVQA val [127]. *The training annotations of the datasets are observed during training. “IViT-6B” represents our InternViT-6B.

Multi-Modal Dialogue Benchmarks

method	visual encoder	glue layer	LLM	Res.	PT	SFT	train. param	image captioning			visual question answering				dialogue	
								COCO	Flickr	NoCaps	VQA ^{v2}	GQA	VizWiz	VQA ^T	MME	POPE
InstructBLIP [34]	EVA-g	QFormer	Vicuna-7B	224	129M	1.2M	188M	–	82.4	123.1	–	49.2	34.5	50.1	–	–
BLIP-2 [81]	EVA-g	QFormer	Vicuna-13B	224	129M	–	188M	–	71.6	103.9	41.0	41.0	19.6	42.5	1293.8	85.3
InstructBLIP [34]	EVA-g	QFormer	Vicuna-13B	224	129M	1.2M	188M	–	82.8	121.9	–	49.5	33.4	50.7	1212.8	78.9
InternVL-Chat (ours)	IViT-6B	QLLaMA	Vicuna-7B	224	1.0B	4.0M	64M	141.4*	89.7	120.5	72.3*	57.7*	44.5	42.1	1298.5	85.2
InternVL-Chat (ours)	IViT-6B	QLLaMA	Vicuna-13B	224	1.0B	4.0M	90M	142.4*	89.9	123.1	71.7*	59.5*	54.0	49.1	1317.2	85.4
Shikra [21]	CLIP-L	Linear	Vicuna-13B	224	600K	5.5M	7B	117.5*	73.9	–	77.4*	–	–	–	–	–
IDEFICS-80B [66]	CLIP-H	Cross-Attn	LLaMA-65B	224	1.6B	–	15B	91.8*	53.7	65.0	60.0	45.2	36.0	30.9	–	–
IDEFICS-80B-I [66]	CLIP-H	Cross-Attn	LLaMA-65B	224	353M	6.7M	15B	117.2*	65.3	104.5	37.4	–	26.0	–	–	–
Qwen-VL [5]	CLIP-G	VL-Adapter	Qwen-7B	448	1.4B [†]	50M [†]	9.6B	–	85.8	121.4	78.8*	59.3*	35.2	63.8	–	–
Qwen-VL-Chat [5]	CLIP-G	VL-Adapter	Qwen-7B	448	1.4B [†]	50M [†]	9.6B	–	81.0	120.2	78.2*	57.5*	38.9	61.5	1487.5	–
LLaVA-1.5 [91]	CLIP-L ₃₃₆	MLP	Vicuna-7B	336	558K	665K	7B	–	–	–	78.5*	62.0*	50.0	58.2	1510.7	85.9
LLaVA-1.5 [91]	CLIP-L ₃₃₆	MLP	Vicuna-13B	336	558K	665K	13B	–	–	–	80.0*	63.3*	53.6	61.3	1531.3	85.9
InternVL-Chat (ours)	IViT-6B	MLP	Vicuna-7B	336	558K	665K	7B	–	–	–	79.3*	62.9*	52.5	57.0	1525.1	86.4
InternVL-Chat (ours)	IViT-6B	MLP	Vicuna-13B	336	558K	665K	13B	–	–	–	80.2*	63.9*	54.6	58.7	1546.9	87.1
InternVL-Chat (ours)	IViT-6B	QLLaMA	Vicuna-13B	336	1.0B	4.0M	13B	146.2*	92.2	126.2	81.2*	66.6*	58.5	61.5	1586.4	87.6

Table 9. **Comparison with SoTA methods on 9 benchmarks.** Image captioning datasets include: COCO Karpathy test [22], Flickr30K Karpathy test [116], NoCaps val [2]. VQA datasets include: VQAv2 test-dev [54], GQA test-balanced [64], VizWiz test-dev [56], and TextVQA val [127]. *The training annotations of the datasets are observed during training. “IViT-6B” represents our InternViT-6B.

Conclusion

- InternVL이라는 대규모 vision-language foundation model을 제안
- 여러 출처의 웹 규모 이미지-텍스트 데이터를 활용해 효율적으로 학습
- 다양한 visual-linguistic tasks 에서의 좋은 성능